SC452 Processes on Networks | Assignment 6 | Question 1 (SI model)

Given: $\beta(t) = \beta_0 e^{-at}$, $S + I = N$, $I(0) = I_0$ with $0 < I_0 < N/2$.

(a) Single equation and exact I(t)

Put $S = N - I$ in the I equation:

$$\frac{dI}{dt} = \beta_0 e^{-at} I\left(1 - \frac{I}{N}\right)$$

Separate variables (partial fractions on the left):

$$\frac{N\,dI}{I(N-I)} = \left(\frac{1}{I} + \frac{1}{N-I}\right)dI = \beta_0 e^{-at} dt$$

Integrate from 0 to t:

$$\ln\frac{I(N-I_0)}{I_0(N-I)} = B(t), \qquad B(t) = \int_0^t \beta_0 e^{-as} ds = \frac{\beta_0}{a}(1 - e^{-at})$$

$$\boxed{I(t) = \frac{N}{1 + \dfrac{N - I_0}{I_0} e^{-B(t)}}}$$

(It is just the logistic curve with t replaced by the 'effective time' B(t).)

(b) Long time limit and the >N/2 condition

As $t \to \infty$, $B(t) \to \beta_0/a$ (finite), so

$$\boxed{I_\infty = \frac{N}{1 + \dfrac{N - I_0}{I_0} e^{-\beta_0/a}} \quad < N}$$

So the infection never reaches everyone. Now $I_\infty > N/2$ iff the denominator is < 2:

$$\frac{N - I_0}{I_0} e^{-\beta_0/a} < 1 \;\Leftrightarrow\; \frac{\beta_0}{a} > \ln\frac{N - I_0}{I_0} \;\Leftrightarrow\; a < \frac{\beta_0}{\ln\left((N - I_0)/I_0\right)}$$

Every step is reversible, so this is necessary and sufficient. Since $I_0 < N/2$ the log is positive.

(c) When does I(t) reach N/2 ?

Let $\Lambda = \ln\frac{N - I_0}{I_0}$. From (a), $I = N/2$ exactly when $B(t) = \Lambda$, i.e.

$$1 - e^{-at} = \frac{a\Lambda}{\beta_0} \quad \Rightarrow \quad t_{1/2} = -\frac{1}{a}\ln\left(1 - \frac{a\Lambda}{\beta_0}\right)$$

- $a\Lambda < \beta_0$: $t_{1/2}$ is finite and positive, N/2 is reached (and crossed) at a finite time.
- $a\Lambda = \beta_0$: $B(t) < \beta_0/a = \Lambda$ for every finite t, so $I(t) < N/2$ but $I \to N/2$ (only asymptotically).

- $a\Lambda > \beta_0$: $B(t)$ never gets to Λ, so N/2 is never reached ($I_\infty < N/2$).

(d) Maximum of the incidence J = dI/dt

Write $x = I/N$, so $J = N\beta x(1-x)$ and $\dot{x} = \beta x(1-x)$, $\dot{\beta} = -a\beta$. Differentiating:

$$\frac{dJ}{dt} = N\left[\dot{\beta}x(1-x) + \beta(1-2x)\dot{x}\right] = N\beta x(1-x)[\beta(1-2x) - a]$$

$N\beta x(1-x) > 0$, so the sign of J' is the sign of $g(t) - a$ with $g(t) = \beta(t)(1-2x)$. Interior max condition:

$$\boxed{\beta_0 e^{-at^*}\left(1 - \frac{2I(t^*)}{N}\right) = a}$$

Max at t = 0: J decreases from the start iff $g(0) \le a$, i.e. $a \ge \beta_0(1 - 2I_0/N)$.

Only one interior max: $g' = -a\beta(1-2x) - 2\beta^2 x(1-x)$. Wherever $g > 0$ (i.e. $x < 1/2$) both terms are negative, so g is strictly decreasing there. Since $a > 0$, g can only hit the level a while $g > 0$, so it crosses a at most once (going downward). Also $g \to 0 < a$ as $t \to \infty$. Hence if $a < \beta_0(1 - 2I_0/N)$ there is exactly one interior maximum, and otherwise none. Two or more interior maxima are not possible.

(e) Numbers: N = 1000, I_0 = 10, β_0 = 2 per day

$$\Lambda = \ln\frac{990}{10} = \ln 99 = 4.5951, \qquad a_c = \frac{\beta_0}{\Lambda} = \frac{2}{4.5951} \approx 0.4352 \text{ day}^{-1}$$

$a < 0.4352$: more than 500 eventually infected (e.g. a = 0.3: $I_\infty \approx 888$, 500 crossed at t = 3.9 d).

$a > 0.4352$: less than 500 eventually infected (e.g. a = 0.6 gives $I_\infty \approx 221$).

$a = a_c$: $e^{-\beta_0/a} = 1/99$, so $I_\infty = 1000/(1+1) = 500$ exactly. But $B(t) < \beta_0/a$ for all finite t, so I(t) stays below 500 and only tends to it as $t \to \infty$ (the middle case of (c)). Since $a_c < \beta_0(1 - 2I_0/N) = 1.96$, the incidence here still has one interior peak before it dies away.